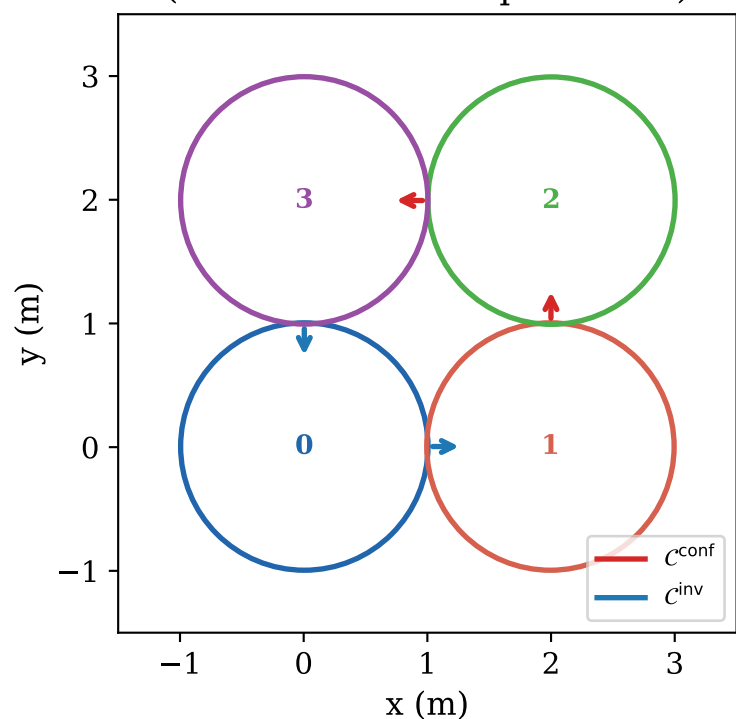
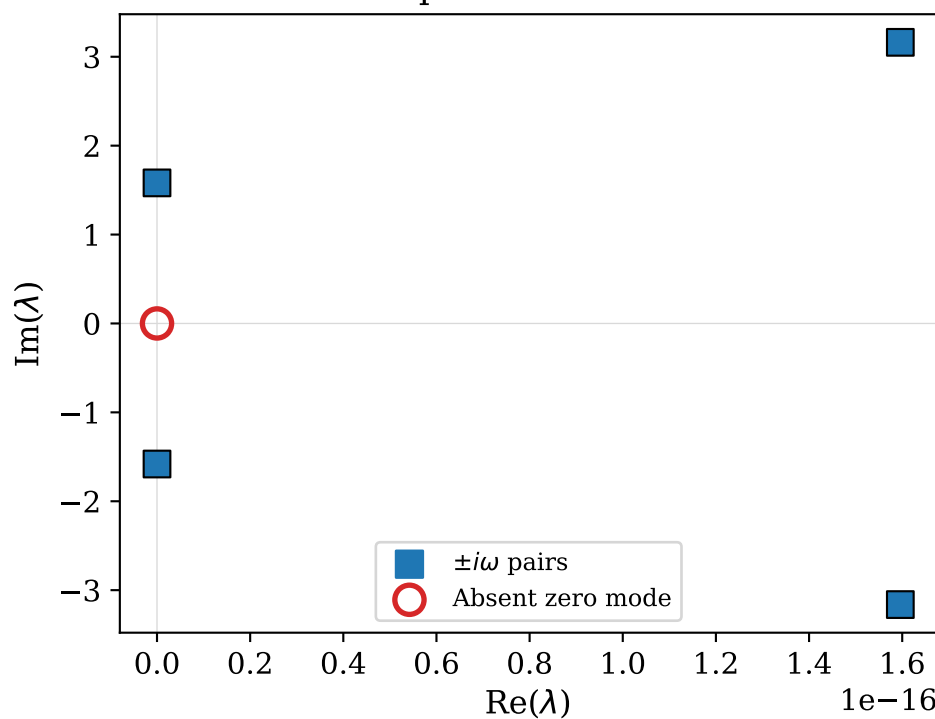


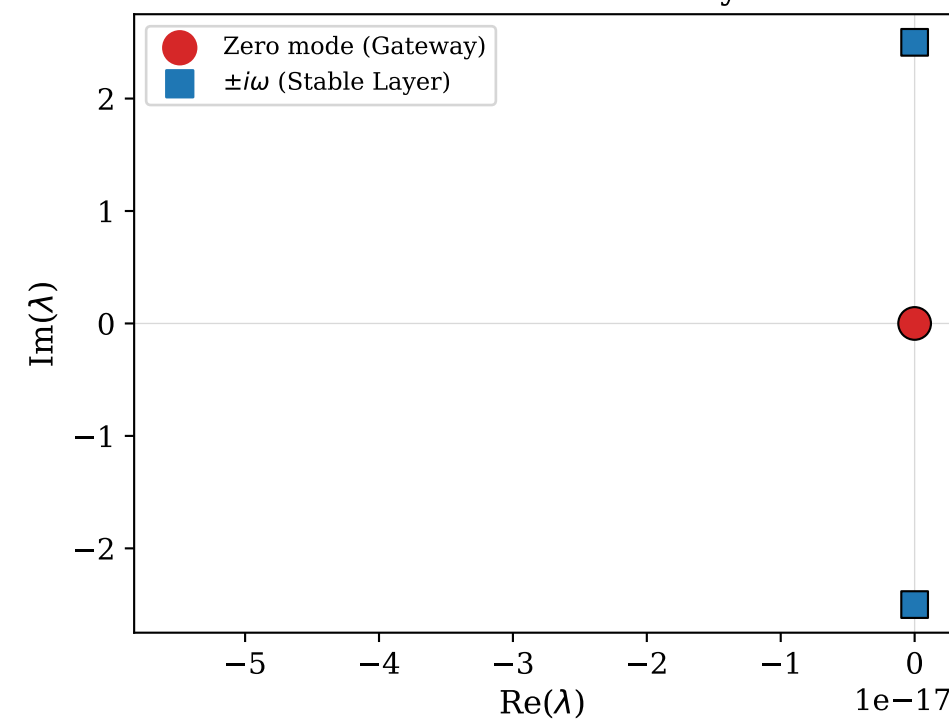
(a) Four-grain square
(initial + loaded + perturbed)



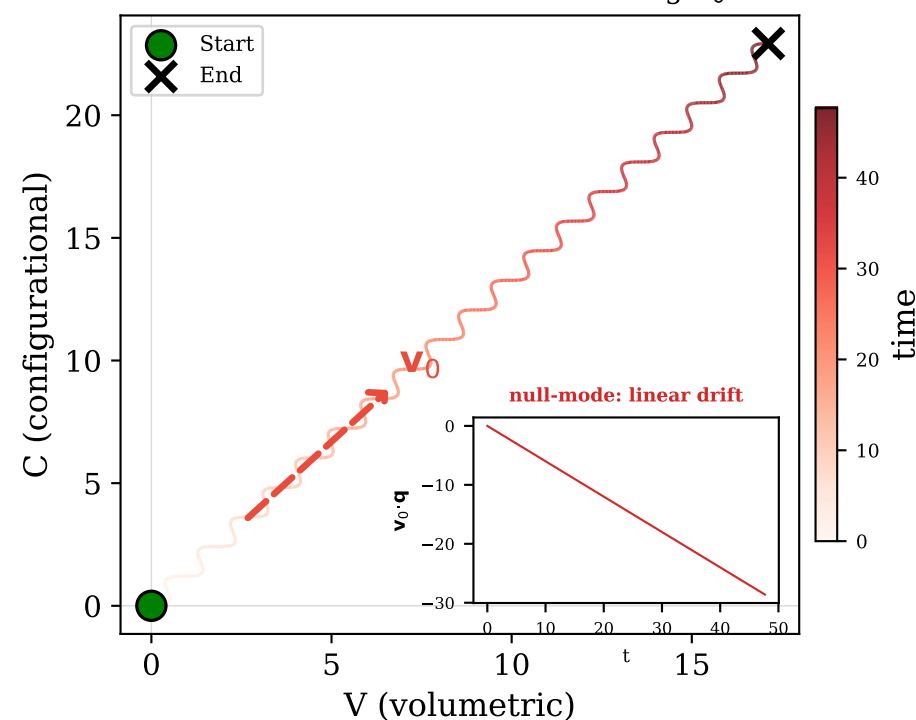
(b) Spectrum $\mathbf{L}^{\text{VMCE}} (N = 4)$
Two $\pm i\omega$ pairs — no zero mode



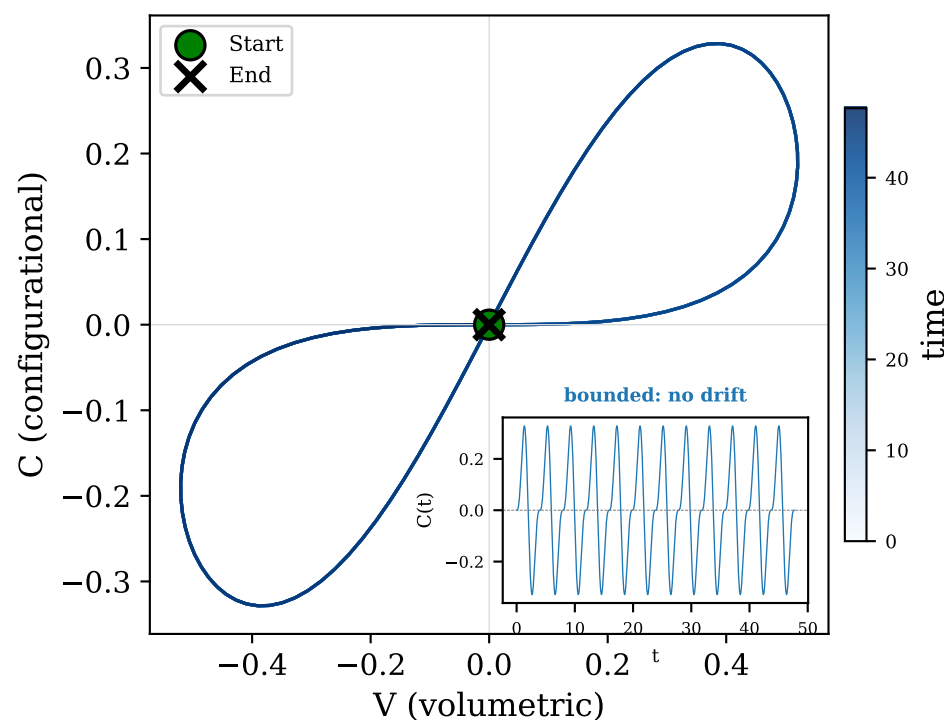
(c) Spectrum $\mathbf{L}^{\text{VMC}} (N = 3)$
Zero mode = Gateway



(d) $N = 3$, pure $\mathbf{L}$
Oscillations + secular drift along $\mathbf{v}_0$



(e) $N = 4$, pure $\mathbf{L}$
Bounded oscillations — no secular drift



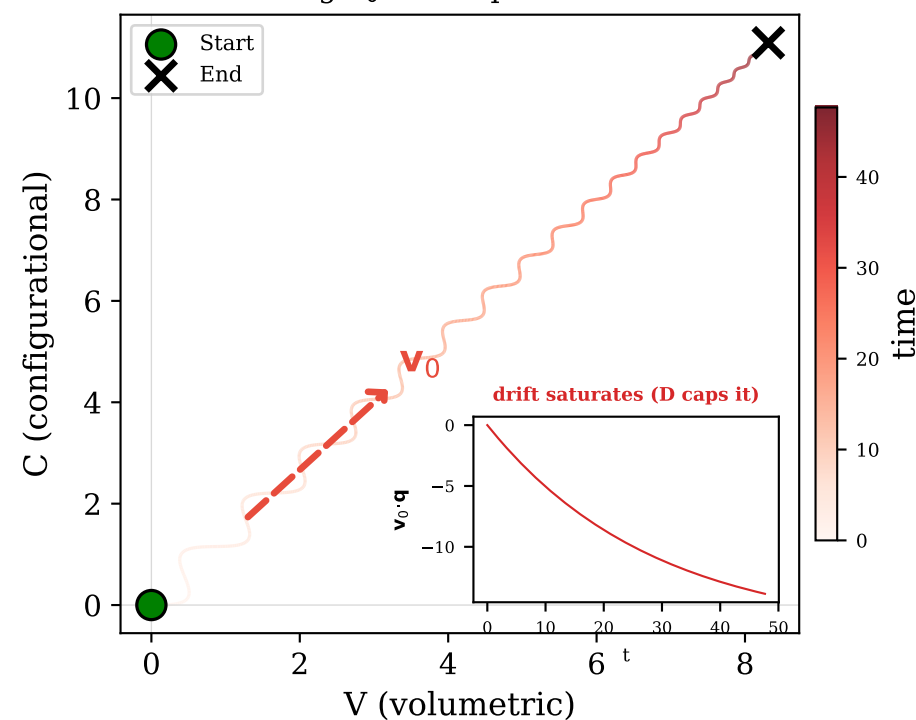
(f) Two tiers of irreversibility

Tier 1: L-channel (topological)
Governed by skew-symmetric $\mathbf{L}$
Zero mode $\rightarrow$ energy frozen in fabric
Parity-controlled: odd N (generically)
 μ -independent

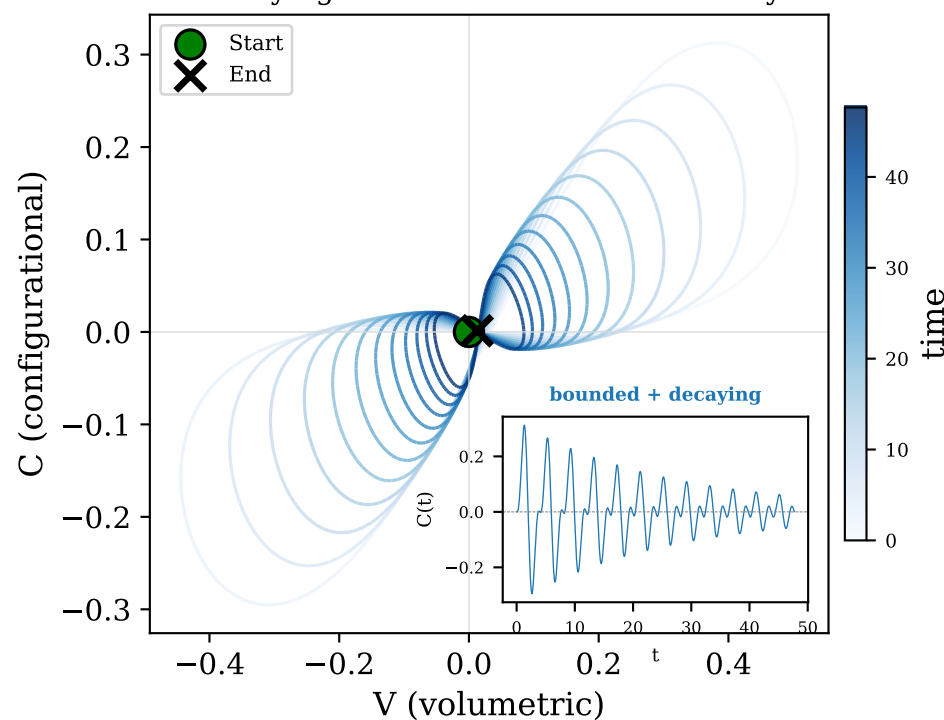
independent at one scale;
connected across scales (Part II)

Tier 2: D-channel (dissipative)
Governed by symmetric $\mathbf{D}$
Friction $\rightarrow$ energy lost to atomic lattice
Always present when $\mu > 0$
Parity-independent

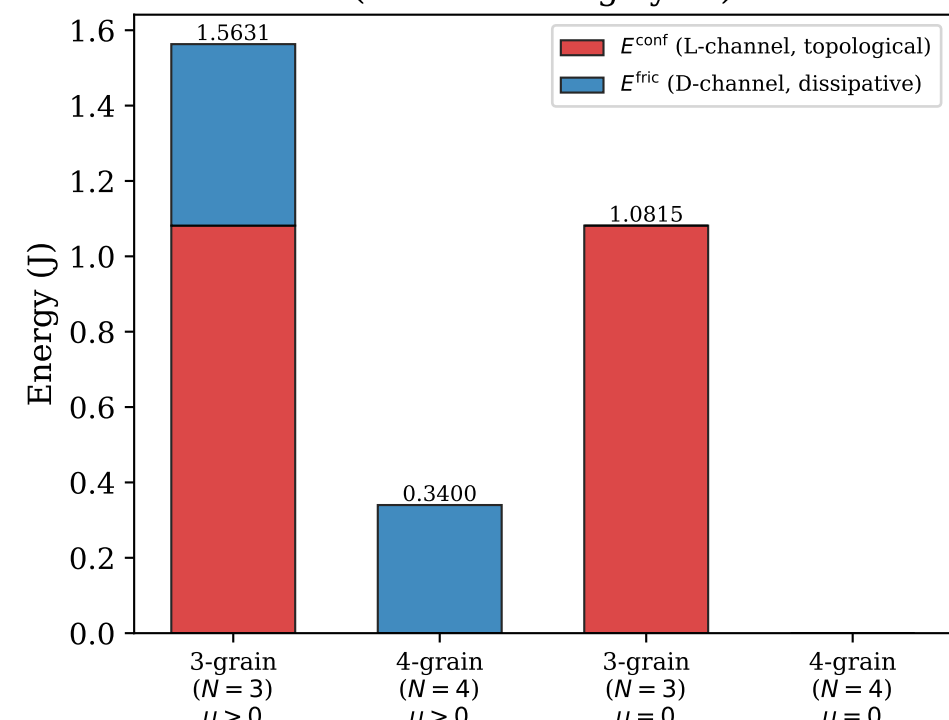
(g) $N = 3$, $\mathbf{L} + \mathbf{D}$
Drift along $\mathbf{v}_0$ + dissipative saturation



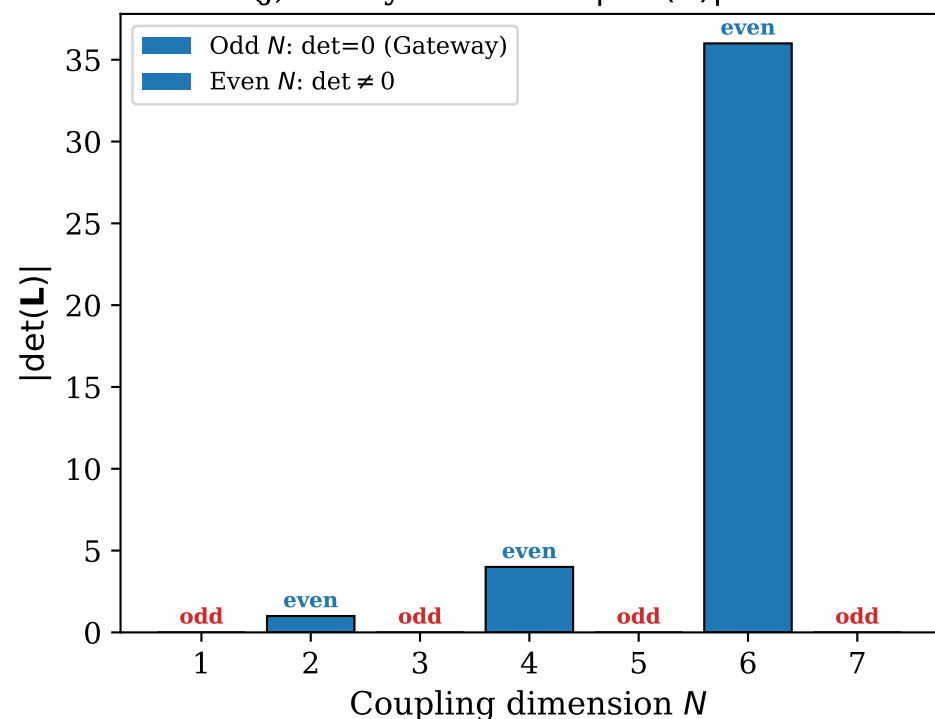
(h) $N = 4$, $\mathbf{L} + \mathbf{D}$
Decaying oscillations — D-channel only



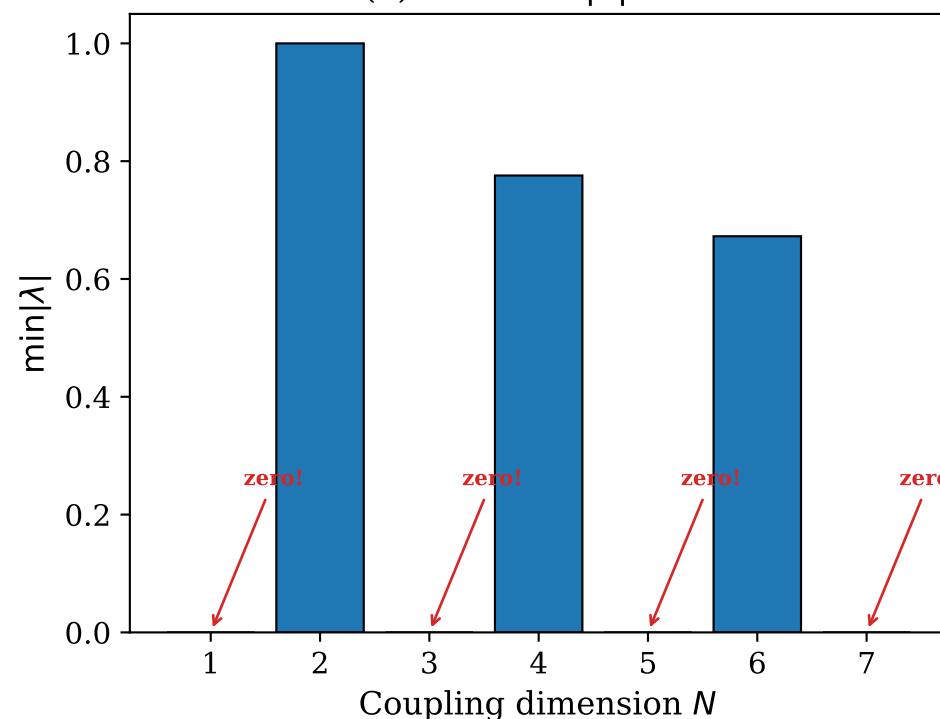
(i) Energy budget: two tiers
(closed loading cycle)



(j) Parity Theorem: $|\det(\mathbf{L})|$ vs N



(k) Smallest $|\lambda|$ vs N



(l) Summary: two-tier comparison

Property	$N = 3$ (odd)	$N = 4$ (even)
$\det(\mathbf{L})$	$= 0$	$= 25.00 = (L_{VM} L_{CE})^2$
Zero mode	YES	NO (generic)
L-channel (config.)	OPEN (topol. irrev.)	GENERALLY CLOSED
D-channel (frictional)	OPEN (always, $\mu > 0$)	OPEN (always, $\mu > 0$)
Pure $\mathbf{L}$ orbits	Drift along $\mathbf{v}_0$	Closed $\pm i\omega$
$\mathbf{L} + \mathbf{D}$ orbits	Drift + saturation	Decay only
$\mu = 0$ irrev.?	YES ($E^{\text{conf}} > 0$)	NO
Novel result	Non-dissipative irrev.	Classical dissip. only